	Raffles Institution Physics Department Revision Package 2	**Forces and Moments**	**2**

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

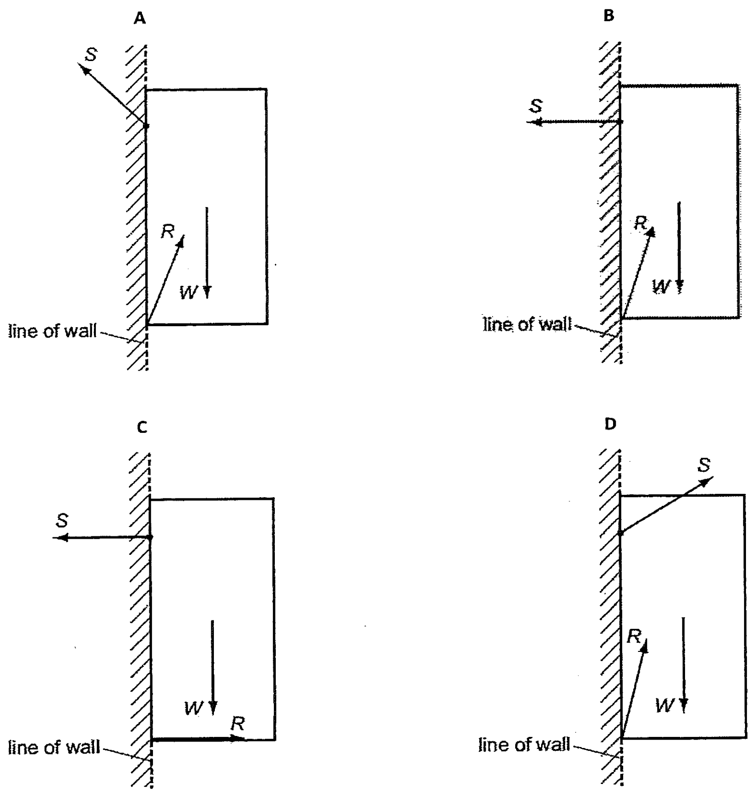
I. MCQs

1. [ACJC/Prelims 2014/P1/8]

A cupboard is attached to a wall by a screw.

Which force diagram shows the cupboard in equilibrium, with the weight W of the cupboard, the force S that the screw exerts on the cupboard and the force R that the wall exerts on the cupboard?

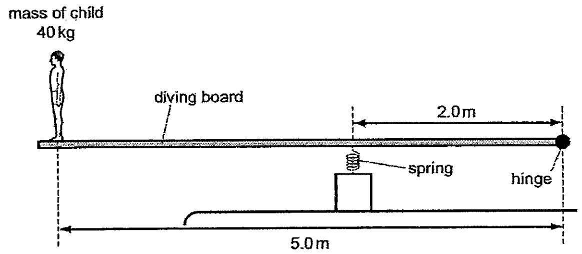
The magnitude of the forces is not drawn to scale.



2. [ACJC/Prelims 2014/P1/10]

A uniform diving board of length 5.0 m and mass 35 kg hangs horizontally in equilibrium when it is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1}.

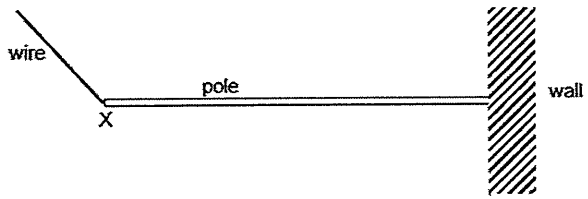
When a child of mass 40 kg stands at the far end of the board as shown in the diagram below, what is the extra compression of the spring caused by the child standing on the end of the board?



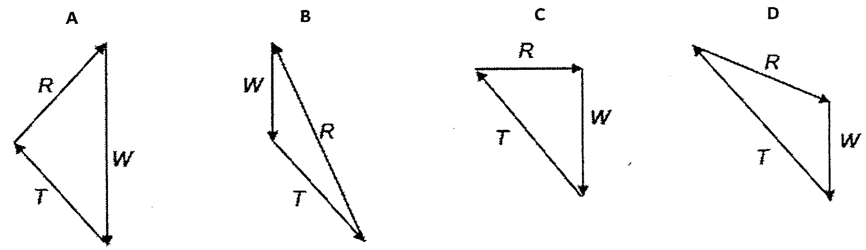
A 1.0 cm B 5.9 cm C 9.8 cm D 19.6 cm

3. [AJC/Prelims 2014/P1/7]

A pole of weight *W* is attached to a wall. It is held horizontal by a wire attached at point X of the pole where *T* is the force of the wire. The pole experiences a reaction force *R* from the wall.

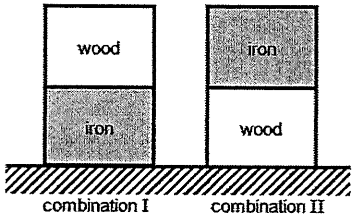


Which triangle of forces could correctly represent the three forces acting on the pole?



4. [CJC/Prelims 2014/P1/5]

Two blocks, one made of wood and the other of iron, are arranged at rest on the ground as depicted in combination I and II below.



Which one of the following statements is correct?

A The force by the iron block on the wooden block in I is greater than that by the wooden block on the iron block in II.

B The force by the wooden block on the iron block in I is the same as that by the iron block on the wooden block in II by virtue of Newton's 3rd law.

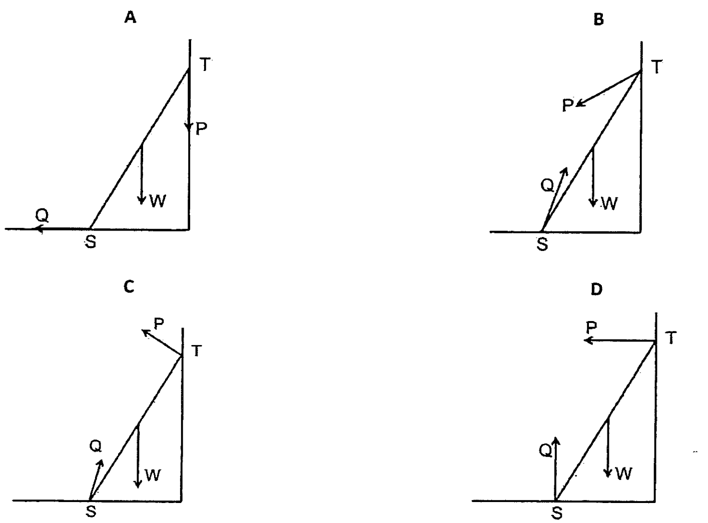
C The force by the wooden block on the iron block is equal to the weight of the wooden block in I while the force by the iron block on the wooden block is equal to the weight of the iron block in II.

D The force by the ground on the iron block in I is greater than the force by the ground on the wooden block in II because the iron block, being denser than the wooden block, exerts more force on the ground.

5. [CJC/Prelims 2014/P1/6]

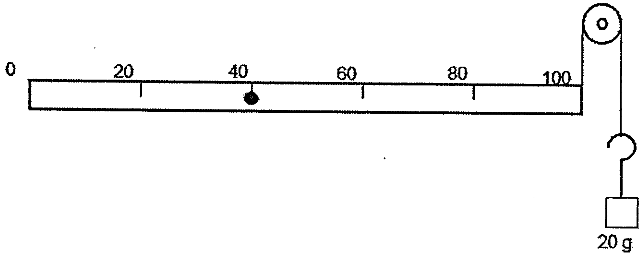
A ladder ST, resting on a rough floor and learning against a rough wall, is on the point of slipping. The ladder has weight W. The contact forces exerted on the ladder by the wall and floor are P and Q respectively.

Which diagram correctly shows the directions of these forces?



6. [DHS/Prelims 2014/P1/7]

A uniform mètre ruler of mass 100 g freely rotates around a pivot at the 40 cm mark. At the 100 cm mark, a string is secured and passed round a frictionless pulley, carrying a mass of 20 g as shown in the diagram.



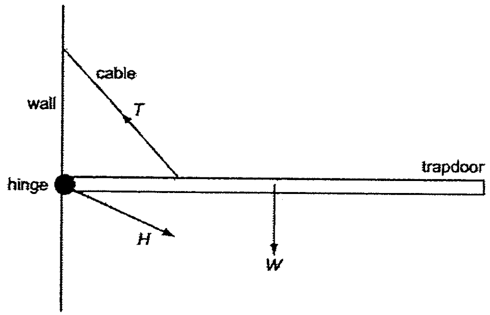
At which mark on the ruler must a 50 g mass be suspended so that the ruler balances?

A 4.0 cm **B** 36 cm **C** 44 cm **D** 96 cm

7. [IJC/Prelims 2014/P1/8]

A hinged trapdoor is held in the horizontal position by a cable.

Three forces act on the trapdoor: the weight W of the trapdoor, the tension T in the cable and the force H at the hinge.



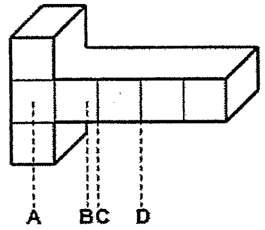
Which list gives the three forces in increasing order of magnitude?

A H, T, W **B** T, H, W **C** W, H, T **D** W, T, H

8. **[JJC/Prelims 2014/P1/5]**

An object of uniform density with uniform cross sectional area is shown in the diagram (drawn to scale).

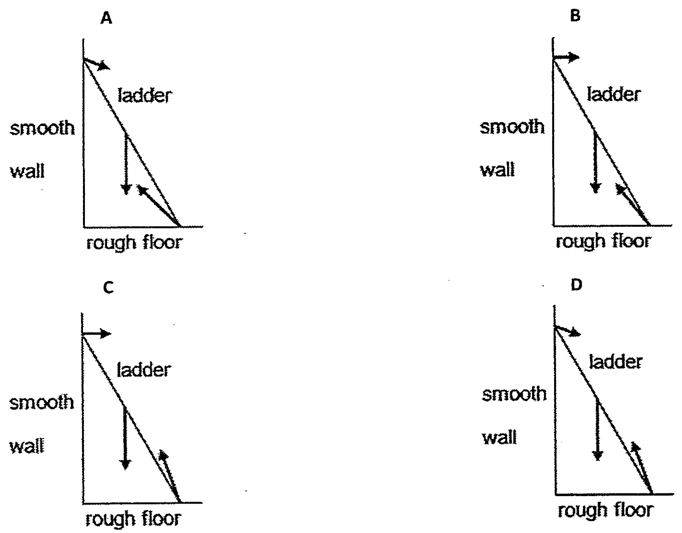
Along which vertical line is its centre of gravity most probably located?



9. **[JJC/Prelims 2014/P1/6]**

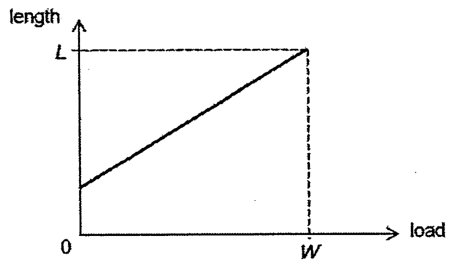
A ladder on a rough floor is leaning against a smooth wall.

Which of the following correctly shows all the forces acting on it?



10. **[MJC/Prelims 2014/P1/7]**

The graph below shows the variation of the length of a spring with the load attached to it. For a load W, the length of the spring is L.



The spring constant of the spring is given by

A the gradient of the graph

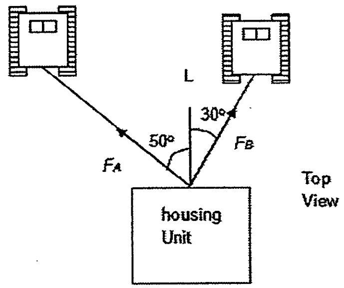
B the inverse of the gradient of the graph

C ratio of W to L

D ratio of L to W

11. **[NJC/Prelims 2014/P1/8]**

Two snow-trucks tow a housing unit at constant speed to a new location in the Antarctica as shown. The sum of forces F_A and F_B exerted on the unit by the horizontal cables is parallel to the line L, and F_A = 4500 N.

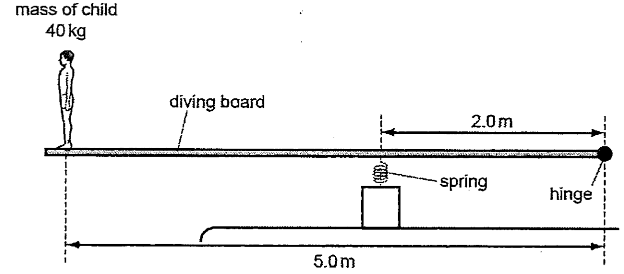


What is F_B?

A 1700 N **B** 2500 N **C** 6900 N **D** 9000 N

12. [NJC/Prelims 2014/P1/9]

A uniform diving board of length 5.0 m and mass 50 kg is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1}. A child of mass 40 kg stands at far end of the board.



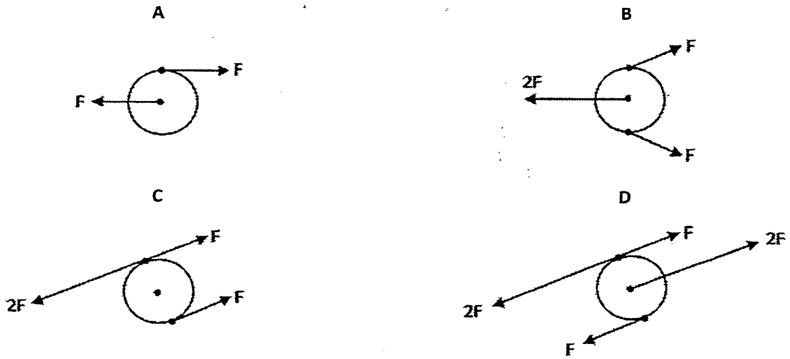
What is the extra compression of the spring caused by the child standing on the end of the board?

A 1.0 cm B 6.1 cm C 9.8 cm D 16 cm

13. [NYJC/Prelims 2014/P1/7]

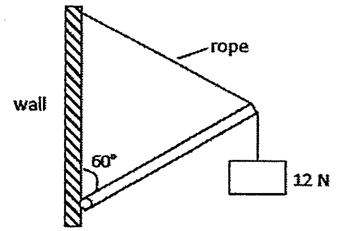
A metal disc is acted upon by a number of forces. The forces are all in the plane of the disc and the weight of the disc is negligible.

In which situation is the disc in static equilibrium?



14. [PJC/Prelims 2014/P1/7]

A uniform rod of weight 20 N and length L is hinged to a wall as shown below. The rod is supported at the other end by a rope of the same length L and attached to the wall. A load of 12 N is suspended at the other end of the rod.

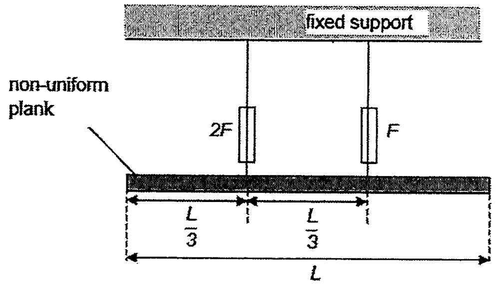


What is the tension in the rope?

A 22 N B 32 N C 57 N D 64 N

15. [RVHS/Prelims 2014/P1/7]

A non-uniform plank of length L is hung horizontally from two spring balances as shown. The forces recorded by the balances are F and $2F$.



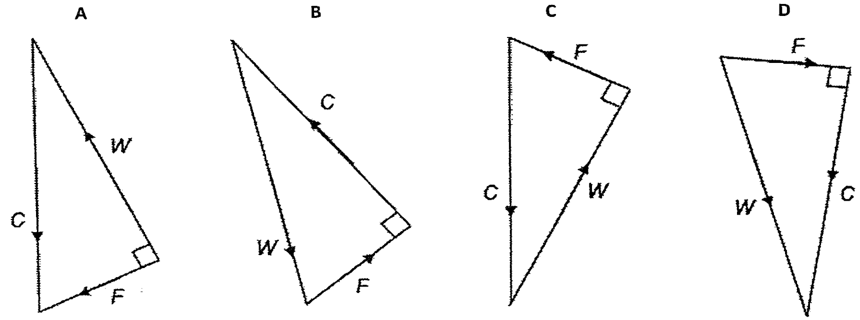
What is the horizontal distance of the centre of gravity of the plank from the right hand end of the plank?

A $\frac{2L}{9}$ B $\frac{4L}{9}$ C $\frac{5L}{9}$ D $\frac{7L}{9}$

16. [SRJC/Prelims 2014/P1/9]

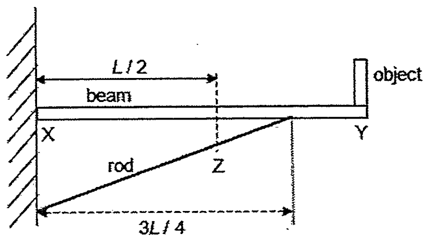
A book slides down a slope at a constant velocity. The three forces that act on the book are the normal contact force *C*, the weight *W* and a constant frictional force *F*.

Which diagram represents these forces acting on the book?



17. [SRJC/Prelims 2014/H1P1/10]

A uniform beam of length *L* is hinged at X and is supported by a rod. An object that has the same mass as the beam is placed at Y as shown.

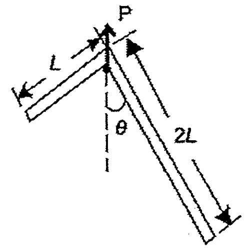


What is the direction of the force exerted on the beam by the wall?

A XZ **B** XY **C** YX **D** ZX

18. [TJC/Prelims 2014/P1/5]

A right-angle rule hangs at rest from a peg P as shown below. It is made from a metal sheet of uniform density. One arm is L cm long while the other is $2L$ cm long.

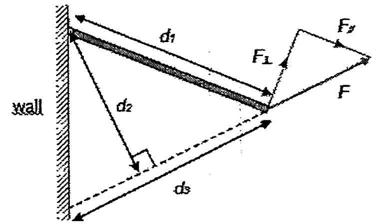


What is the angle θ at which it will hang?

A 8.0° **B** 14° **C** 42° **D** 76°

19. [TPJC/Prelims 2014/P1/4]

A rod is hinged to a wall as shown. A force *F* is applied to one end of the rod. $F_\perp$ and $F_{//}$ are components of *F* which are perpendicular and parallel to the rod respectively.

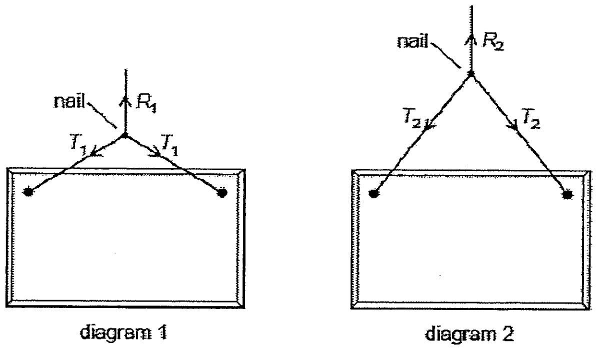


What is the moment due to the force F about the hinge?

A $F d_2$ **B** $F d_3$ **C** $F_{//} d_1$ **D** $F_\perp d_2$

20. [TPJC/Prelims 2014/P1/5]

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

A $R_1 = R_2 \quad T_1 < T_2$

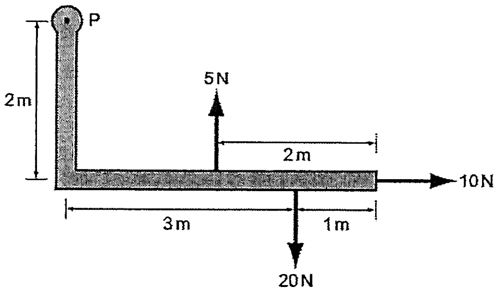
B $R_1 = R_2 \quad T_1 > T_2$

C $R_1 > R_2 \quad T_1 < T_2$

D $R_1 < R_2 \quad T_1 = T_2$

21. [VJC/Prelims 2014/P1/7]

An L-shaped rigid lever arm is pivoted at point P.



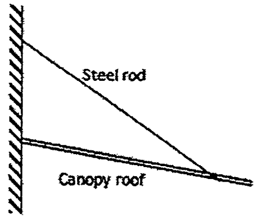
Three forces act on the lever arm, as shown in the diagram.

What is the magnitude of the resultant moment of these forces about point P?

A 30 N m **B** 35 N m **C** 50 N m **D** 90 N m

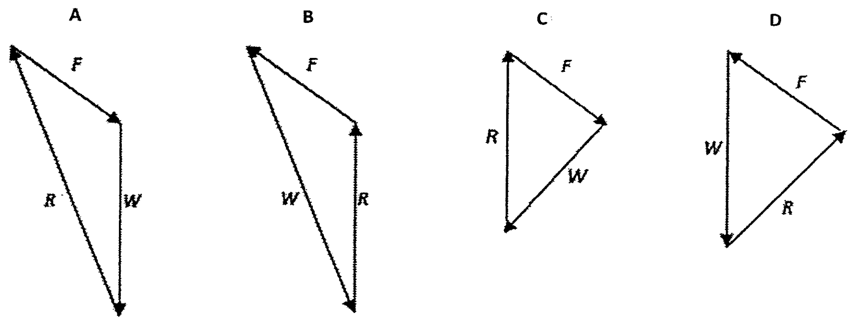
22. [VJC/Prelims 2014/P1/8]

A canopy roof, hinged to a vertical wall at one end and secured by a steel rod at the other end as shown, is in equilibrium.



The weight of the canopy roof is W, the force exerted by the rod on the roof is F and the reaction by the wall on the roof is R.

Which vector triangle represents the forces acting on the canopy roof?



II. Structured Questions

1. **[HCI/Prelims 2014/P2/1]**

   a. State the principle of moments. [1]

   b. A uniform rigid rod of mass 30 kg is attached to a vertical wall by a hinge as shown in Fig. 2.1. The other end of the rod is held to the ceiling by a cable.

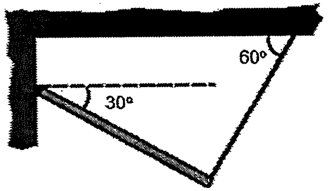


Fig. 1.1

   i. Draw the free body diagram of the forces acting on the rod in Fig. 1.1. Label and explain all the forces clearly. [2]

   ii. Show that the tension T in the cable is 127 N. [2]

   iii. Determine the force acting on the rod by the hinge. [4]

[Total: 9]

2. **[NYJC/Prelims 2014/P2/3]**

A uniform sheet of steel weighing 800 N is supported by a bolt at its lower-left hand corner and by a cable tied to a point on its left-edge as shown in Fig. 2.1 below. The pull by the cable on the sheet is T.

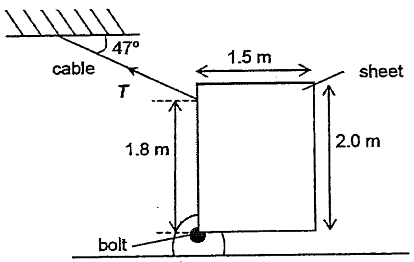


Fig. 2.1

   a. Show that T is 489 N [2]

   b. Determine the magnitude of force acting on the bolt by the sheet. [3]

[Total: 5]

3. **[SRJC/Prelims 2014/P2/3]**

As shown in Fig 3.1, a load of mass 23 kg is placed at 0.30 m from end A of a uniform plank AB. The plank has a weight of 80 N and a length of 2.00 m. Two blocks, block 1 and block 2, are placed beneath the plank to support it. The plank is in equilibrium.

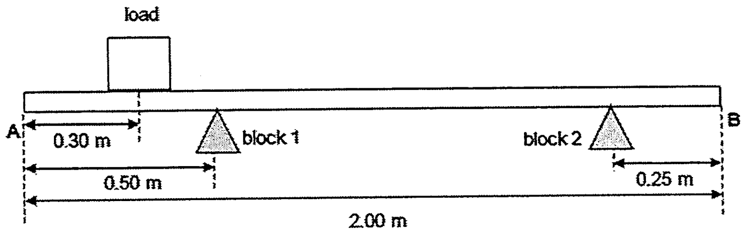


Fig. 3.1

   a. State the two conditions necessary for the plank to be in equilibrium. [2]

   b. The load is moved along the plank, determine whether it is possible for the plank to lose contact with block 1 as the load is moving. [3]

[Total: 5]

	Raffles Institution **Physics Department** **Revision Package 2**	**Forces and Moments**	**2**

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	C	A	C	C	C	C	B	C	B	C	C	D	A	C

16	17	18	19	20	21	22
B	C	B	A	B	A	D

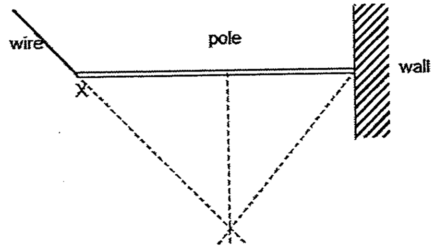
1 A No resultant force. The lines of action of the three forces must pass through a common point.

2 C additional clockwise moment due to spring = additional anti-clockwise moments due to weight of child

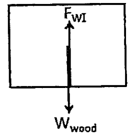
$$(10\times10^3\times e)(2.0)=(40)g(5.0)$$
$$e=0.0981\text{ m}$$

3 A The lines of action of the three forces must pass through a common point.

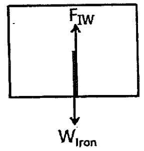
Hence direction of R must act along the dotted line from the wall.



4 C Consider FBD of the wooden block in I. Since it is at equilibrium, its weight is equal to the normal contact force on the wooden block by the iron block for I. (Note: Both are not an action-reaction pair)



Consider FBD of the iron block in II, since it is at equilibrium, its weight is equal to the normal contact force on the iron block by the wooden block for II. (Note: Both are not an action-reaction pair)



Note: Weight of wooden block is LESSER than weight of iron block.

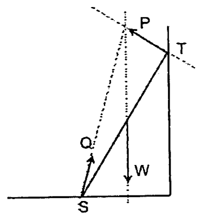
A is wrong as force by iron block on wooden block in I should be LESSER not greater.

B is wrong as force by wooden block on the iron block in I is LESSER than force by iron block on the wooden block in II.

For D, treat the iron and wooden block as one object. At equilibrium, their combined weight is equal to the force by the ground on the iron block for I. At equilibrium, their combined weight is equal to the force by the ground on the wooden block for II. The statement should be the force by the ground on the iron block in I is the SAME as the force by the ground on the wooden block in II since their combined weights are the same for both scenarios.

5 C All the lines of action of forces acting on the ladder intersect at a single point when it is in equilibrium.

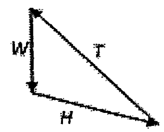
For option B, though the lines of action of the forces act at a single point, the force P is made up of both the normal and frictional forces and the frictional force should act upwards.



6 C Taking moment about the pivot,

$$\text{clockwise moment} = \text{anticlockwise moment}$$
$$(0.050)g\times(r-40)+(0.100)g\times(50-40)=(0.020)g(100-40)$$
$$r=44\text{ cm}$$

7 C Sketch a force diagram with the 3 forces forming a closed triangle.



8 B A and D obviously wrong.
Let x = length of each cube and W = weight of each cube
Try taking moments about C,
clockwise moment = $(3W)(1.5x) = 4.5Wx$
anti-clockwise moment = $(3W)(1.5x) + (W)(0.5x) = 4.5Wx + 0.5Wx = 5\ Wx$
Therefore C is also wrong and centre of gravity is slightly to the left side of C.

9 C Smooth wall means no frictional force, only normal reaction from wall, hence A and D are wrong.
For option B, by taking moment about c.g., the contact forces on ladder by smooth wall and rough floor will produce resultant clockwise moment, so B is wrong.

10 B Recall: To get spring constant, we need the gradient of load (Y-axis) against extension of spring (X-axis).
Note that the graph is length of spring (Y-axis) against load (X-axis).
Hence, from the graph given, the inverse of the gradient of the line would be the spring constant.

11 C $(4500)\sin 50° = F_B \sin 30°$
$F_B = 6900$ N

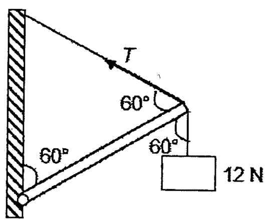
12 C Without child, taking moment about the hinge,
clockwise moment = anticlockwise moment
$(kx_1)(2.0) = (50)g(2.5)$
$x_1 = 0.0613$ m
With child, taking moment about the hinge,
clockwise moment = anticlockwise moment
$(kx_2)(2.0) = (40)g(5.0) + (50)g(2.5)$
$x_2 = 0.159$ m
Extra compression = 0.159 – 0.0613 = 0.0981 m

13 D In option D, the clockwise moment produced by the pair of F forces are balanced by the anticlockwise moment produced by the pair of $2F$ forces.

14 A Taking moments about the hinge,

$$T\sin 60°(L) = 20\sin 60°\left(\frac{L}{2}\right) + 12\sin 60°(L)$$

$T = 22$ N

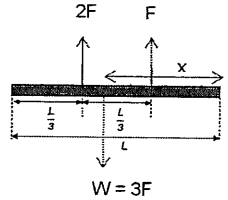


15 C Take moments about the right end,

$$Wx = 2F\left(\frac{2L}{3}\right) + F\left(\frac{L}{3}\right)$$

$$3Fx = 2F\left(\frac{2L}{3}\right) + F\left(\frac{L}{3}\right)$$

$$x = \frac{5L}{9}$$



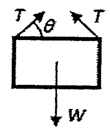
16 B Since object slides down at constant velocity, the object is in equilibrium (acceleration = 0 m s^{-2}). Hence, the force triangle is either B or C.
The angle between the frictional force and the normal contact force is 90°. Hence, the answer is B.

17 C Since the object and beam has the same mass, the combined CG of the object and beam is at 3L/4. As the rod is under compression and by concurrent forces, the direction of force exerted on the beam by the wall is YX.

18 B ACW moments = CW moments
$2m \times (L \sin\theta) = m \times [\frac{1}{2} L \sin(90 - \theta)]$
$\tan\theta = \frac{1}{4} \Rightarrow \theta = 14°$

19 A Moment = force × perpendicular distance from the line of action of the force = $F\,d_2$
Or
Moment = distance from pivot to the force × perpendicular component of the force = $F_{\perp}d_1$
Since this is not in the options, **A** is the answer.

20 B Since it is the same picture, the weight of the picture is constant and $R_1 = R_2$.
$2T\sin\theta = mg$
For diagram 1, θ is smaller and $\sin\theta$ will be smaller.
Hence, T_1 will be larger.



21 A Total *clockwise* moments, 20 × 3 = 60 N m
Total *anticlockwise* moments, (5×2) + (10×2) = 30 N m
Resultant moments, 60 – 30 = 30 N m

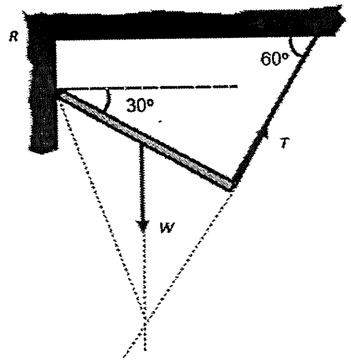
22 D For *equilibrium*, the *lines of action* of the three forces must pass through a common point.

Structured Questions

1. **[HCI/Prelims 2014/P2/1]**

   a. See definitions in notes.

   b. i.



W is the weight of the rod, T is the tension acting on the rod, R is the force acting on the rod by the hinge.

   ii. Take moments about the hinge,

$$30\times 9.81\times \cos 30^\circ\left(\frac{L}{2}\right)=T\times L$$
$$T=127N$$

   iii. Taking towards the left as positive, $R_X = T\sin 30^\circ = 127.4\sin 30^\circ = 63.7N$

Taking upwards as positive, $R_Y = 30\times 9.81 - T\cos 30^\circ = 184N$

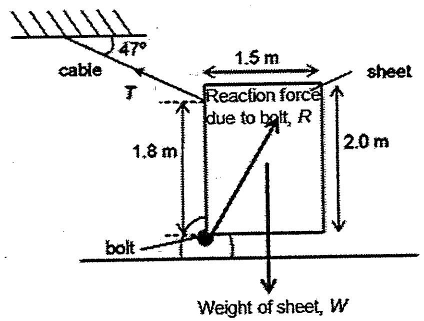
$$R=\sqrt{R_X^2+R_Y^2}=\sqrt{63.7^2+184^2}=195N$$

$\tan\theta=\dfrac{184}{63.7}$ $\Rightarrow$ $\theta = 70.9^\circ$ above the horizontal as shown above

2. **[NYJC/Prelims 2014/P2/3]**

   a. Taking moments about the bolt,

$$Wd_1 = T_x d_2$$
$$800(1.5/2)=(T\cos 47)(1.8)$$
$$T=489\text{ N}$$



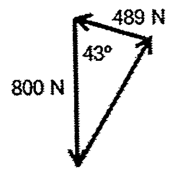
   b. Using Cosine Rule,

$$R=\sqrt{W^2+T^2-2WT\cos\theta}$$
$$R=\sqrt{800^2+489^2-2(800)(489)\cos 43}$$

Force on sheet by bolt, $R = 554$ N

By Newton's third Law, force on bolt by sheet is equal in magnitude but opposite in direction to force on sheet by bolt.

Hence, force on bolt by sheet = 554 N



3. **[SRJC/Prelims 2014/P2/3]**

   a. See definitions in notes.

   b. When load is at distance x to the right of block 2 and plank loses contact with block 1, taking moments about block 2,

$$80\times 0.75 = 23\times 9.81\times x \quad \text{(when lose contact, no force due to block 1)}$$
$$x = 0.266\text{ m}$$

Since $x > 0.25$ m, the plank will not lose contact with block 1.

Note: When plank loses contact with block 1 and remains in contact with block 2, load should move to the right of block 2.